

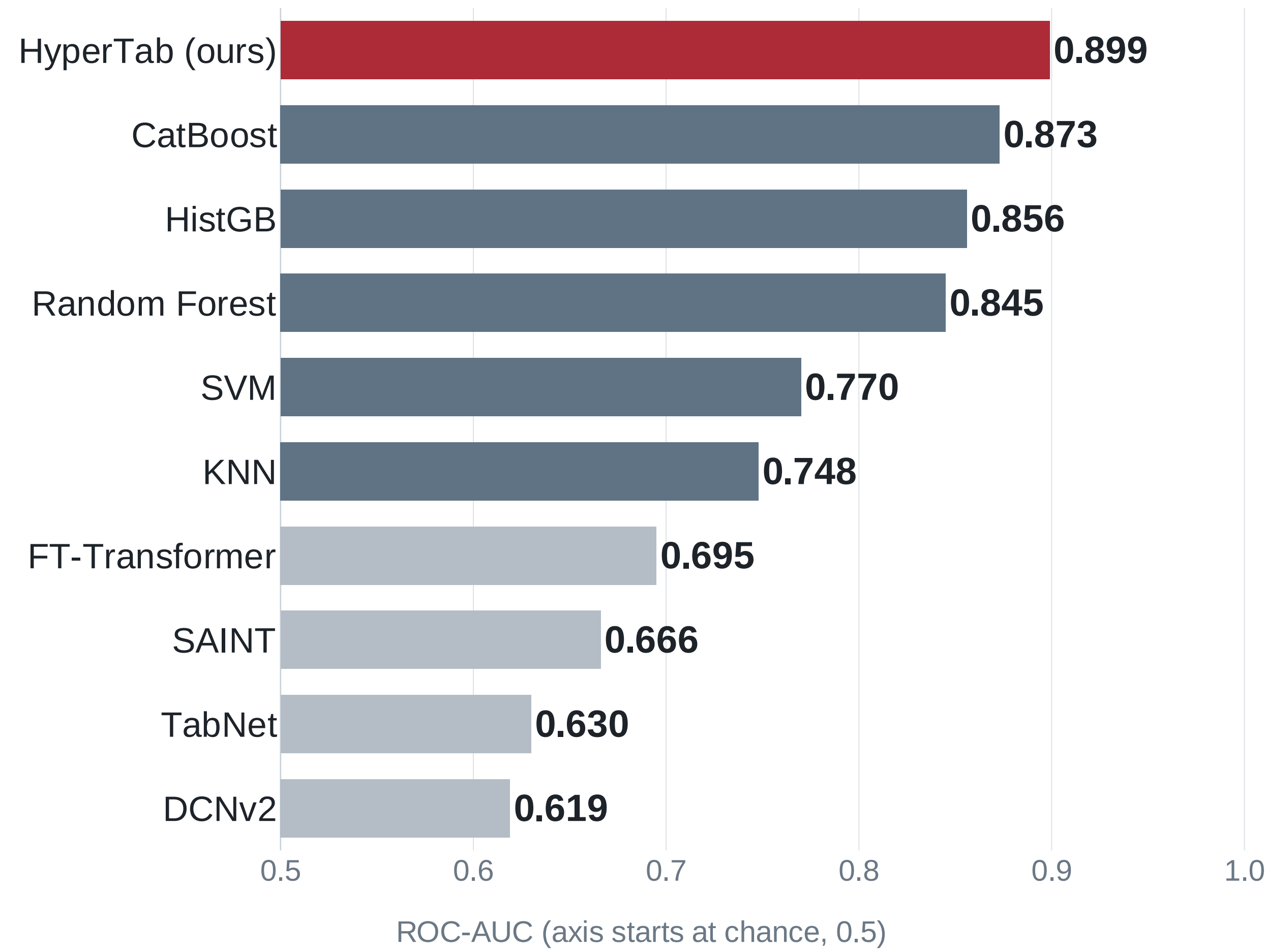


HyperTab, our in-context TabPFN framework, ranks first of 14 models for hypertension screening from voice and demographics.
Demographics carry most of the signal; voice alone is weak but real.

1 HyperTab ranks first of 14 models

0.899 ROC-AUC (± 0.088)
PR-AUC 0.970 (± 0.020)

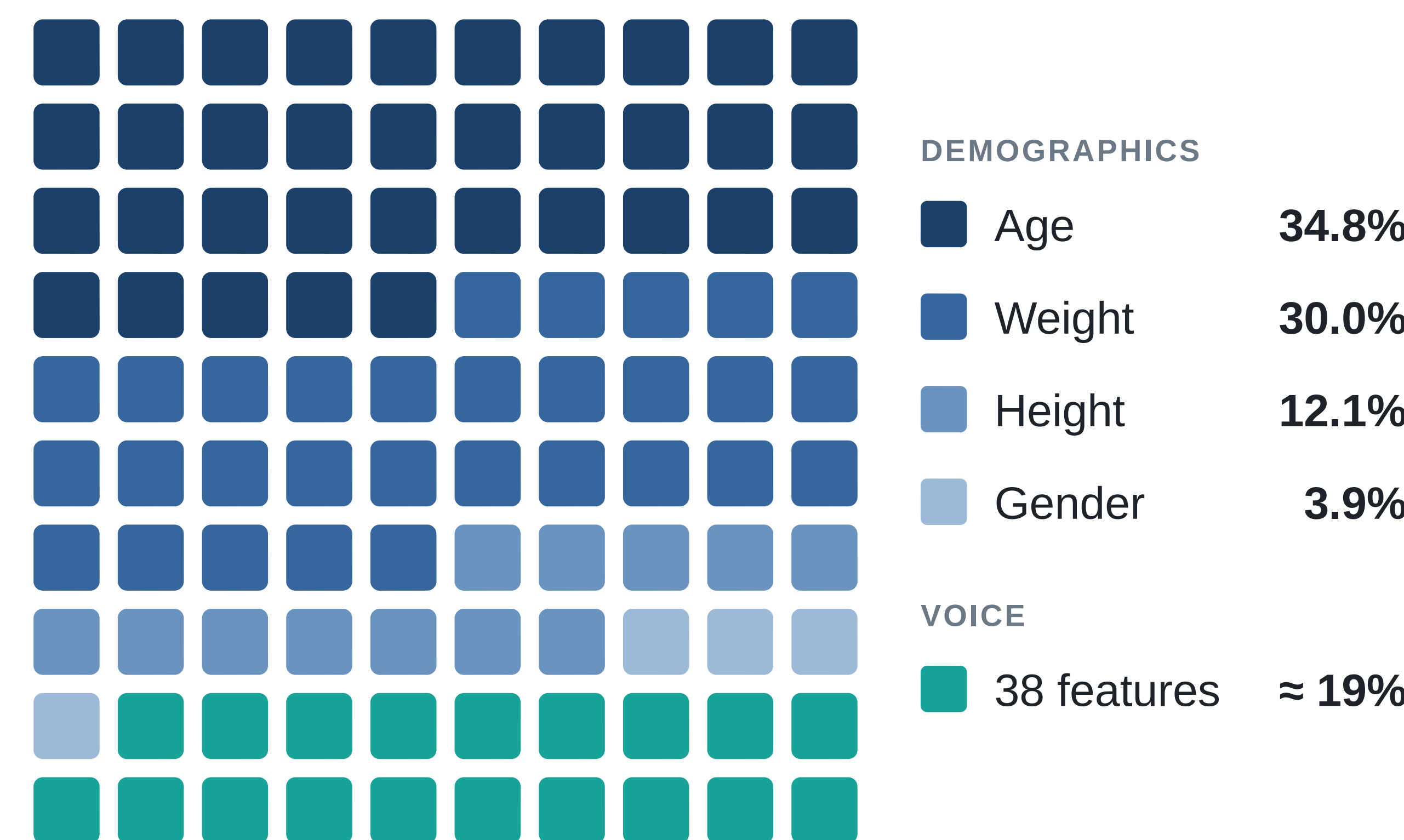
■ HyperTab (ours) ■ Classical ML ■ Deep tabular, from scratch



All 42 features; 10 of the 14 models shown. The gap to CatBoost (0.873) is not statistically significant.

2 Demographics dominate the signal

$\approx 81\%$ of feature importance is demographic



Demographics alone match the full model (ROC-AUC)

0.913
demographics only

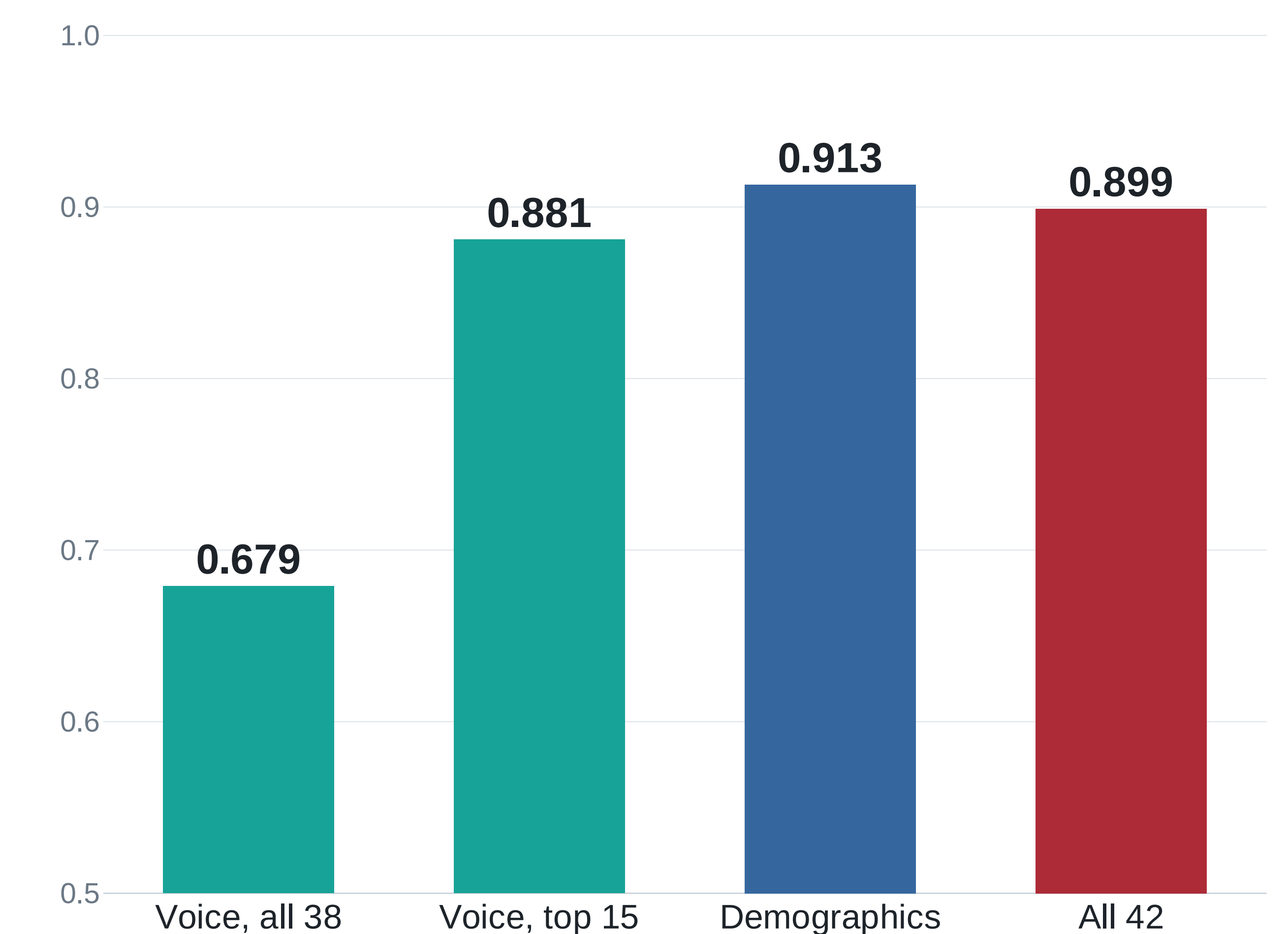
0.899
all 42 features

CatBoost feature importance, all 42 features; one square = 1%. Top voice features: max and peak amplitude (8.2% each).

3 Voice alone is weak but real

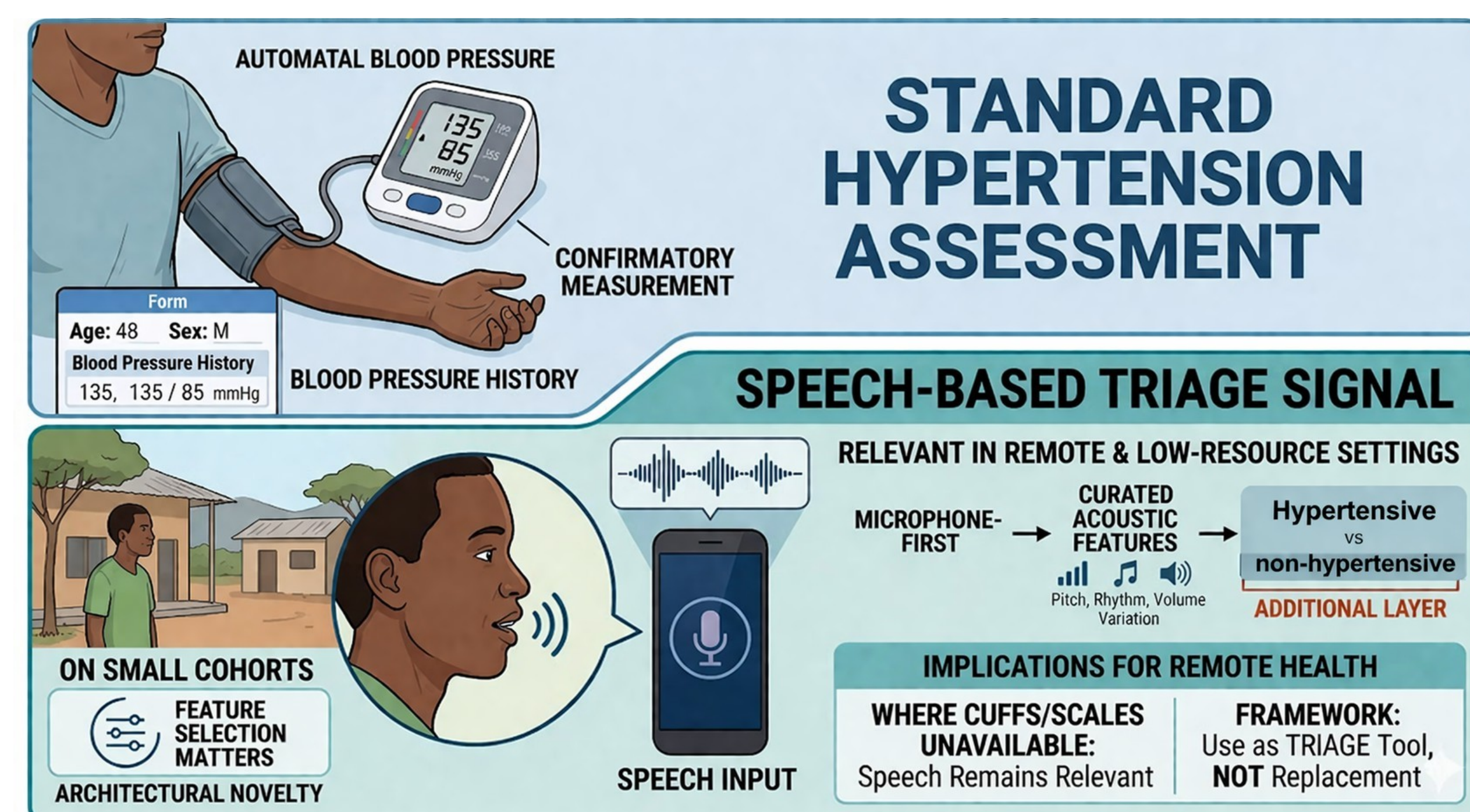
0.881 ROC-AUC from voice alone
top 15 features (all 38: 0.679)

■ Voice ■ Demographics ■ Both (full model)



HyperTab ROC-AUC by input features; the axis starts at chance (0.5). Top 15: voice features kept by feature selection.

Why voice?



1.3 billion

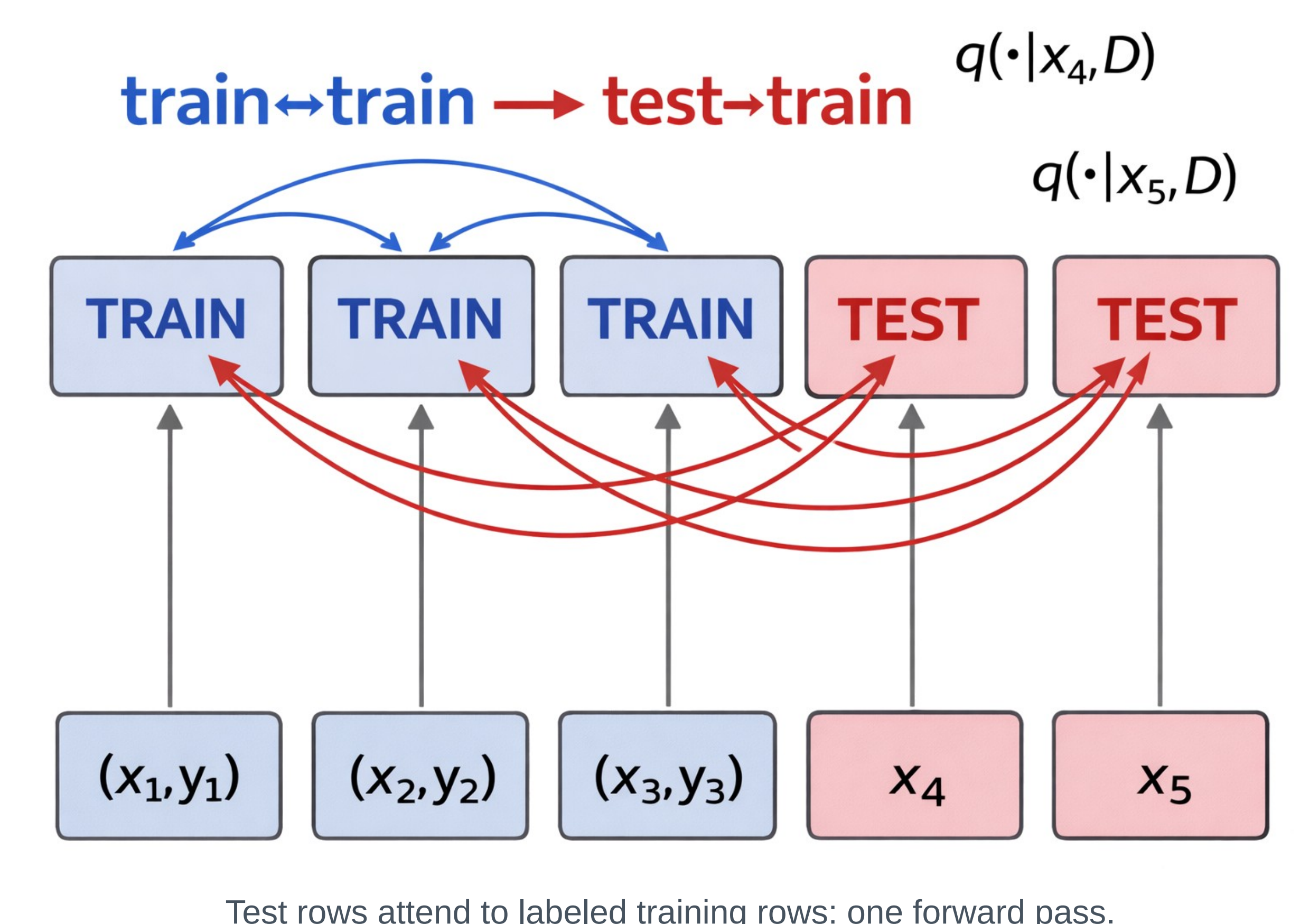
adults have hypertension [1]

44%

are unaware of it [1]

How HyperTab works

- Record sustained vowels**
108 subjects, 58 hypertensive ($\geq 130/80$ mmHg) [2]
- Extract features**
5,380 windows of 50 ms: 38 voice features + 4 demographics
- Split by subject**
Leakage-aware 5-fold CV: train and test never share a subject
- Predict in context (TabPFN)**
Pretrained on synthetic tables; no gradient updates [3, 4]
- Decide per subject**
Top-K mean of window scores, then a threshold



A pretrained tabular prior is **at least on par with tuned boosting** on a small cohort.

Voice alone is **at best a coarse triage signal**, not a replacement for clinical assessment.

Limits: 108 young subjects, one country. **Next:** raw audio, larger cohorts, external validation.